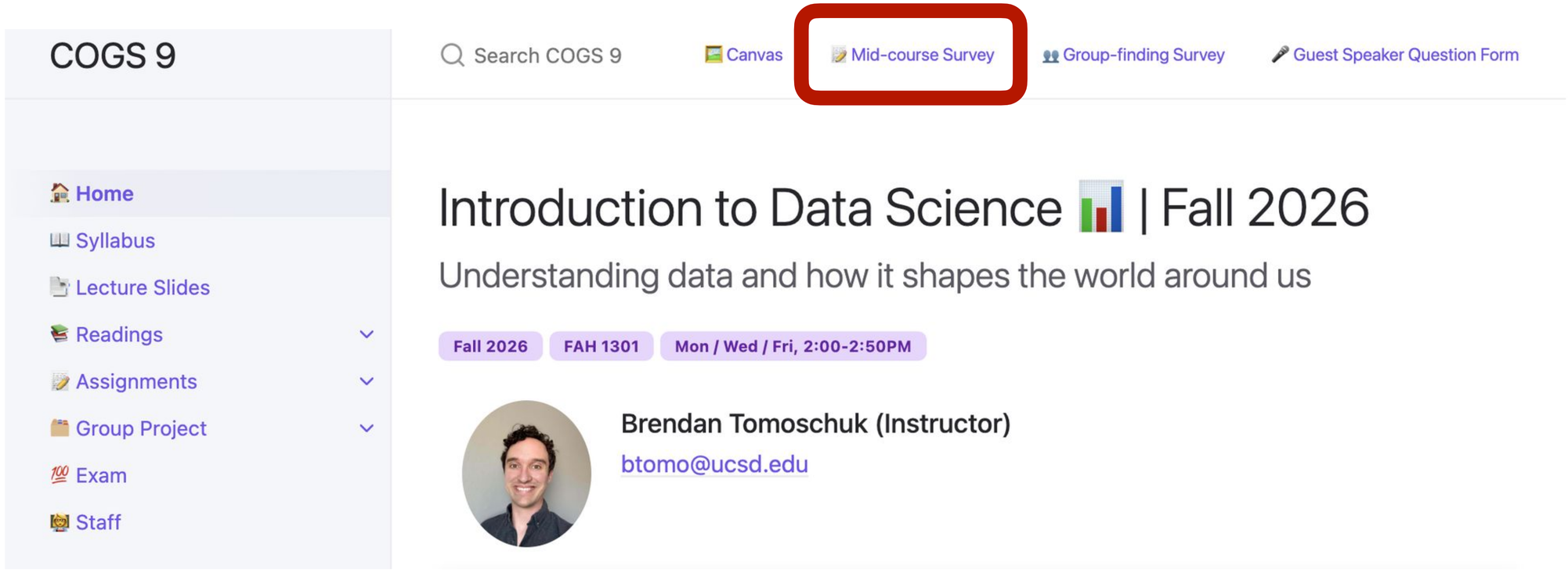
Mid-quarter survey (optional)

- I want this class to be educational, engaging and enjoyable
- Let me know what is going well and what could be better to improve the rest of the quarter, and future classes



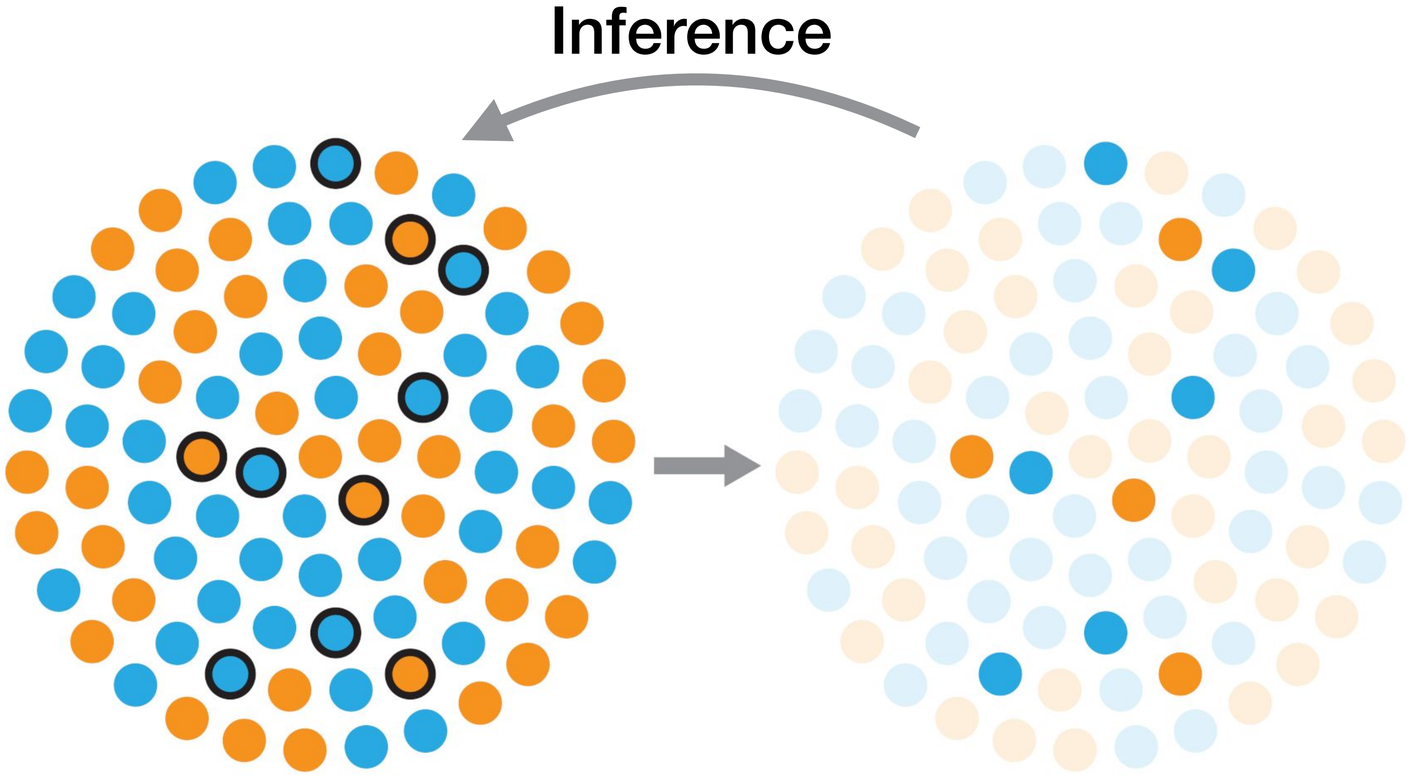
cogs 9 | Week 5

EDA & Statistical Testing

Brendan Tomoschuk

Statistical Inference

- Statistical inference is the process of drawing conclusions about a population from a sample of data



Population

All students in COGS 9

Sample

A subset of students in COGS 9

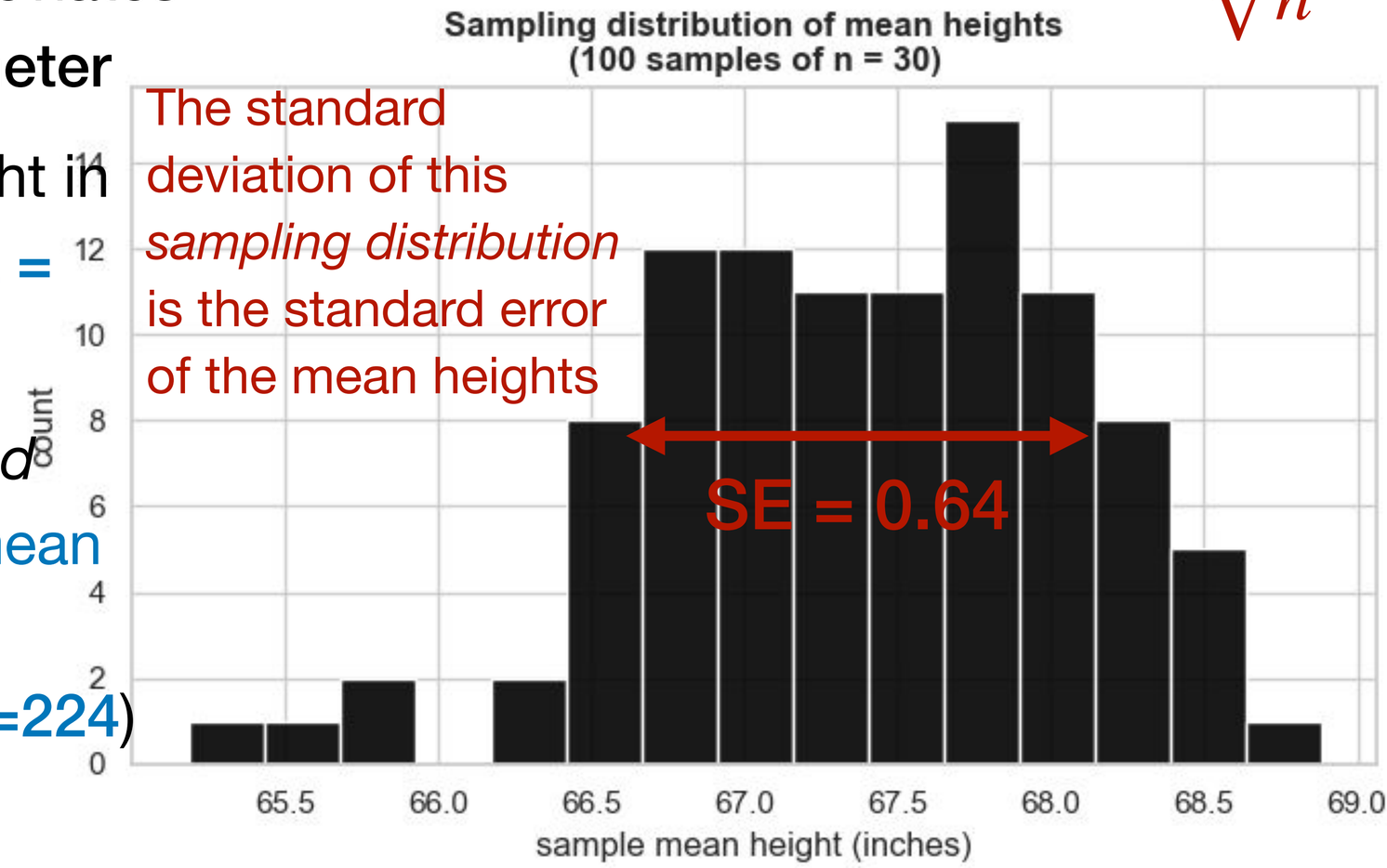
Types of statistical inference

- **Univariate inference:** estimate a single population parameter (e.g., μ, σ) about a single variable
- **Bivariate inference:** draw conclusions about the relationship between two variables (comparison or association)
  - Quantitative ↔ quantitative: test association in values
  - Categorical ↔ quantitative: difference in means across groups
  - Categorical ↔ categorical: test association or difference in proportions
- **Multivariate inference:** drawing conclusions about relationships between variables while accounting for other variables; detect confounding and interaction effects *(study this in a later course)*

Quantifying uncertainty of a sample statistic

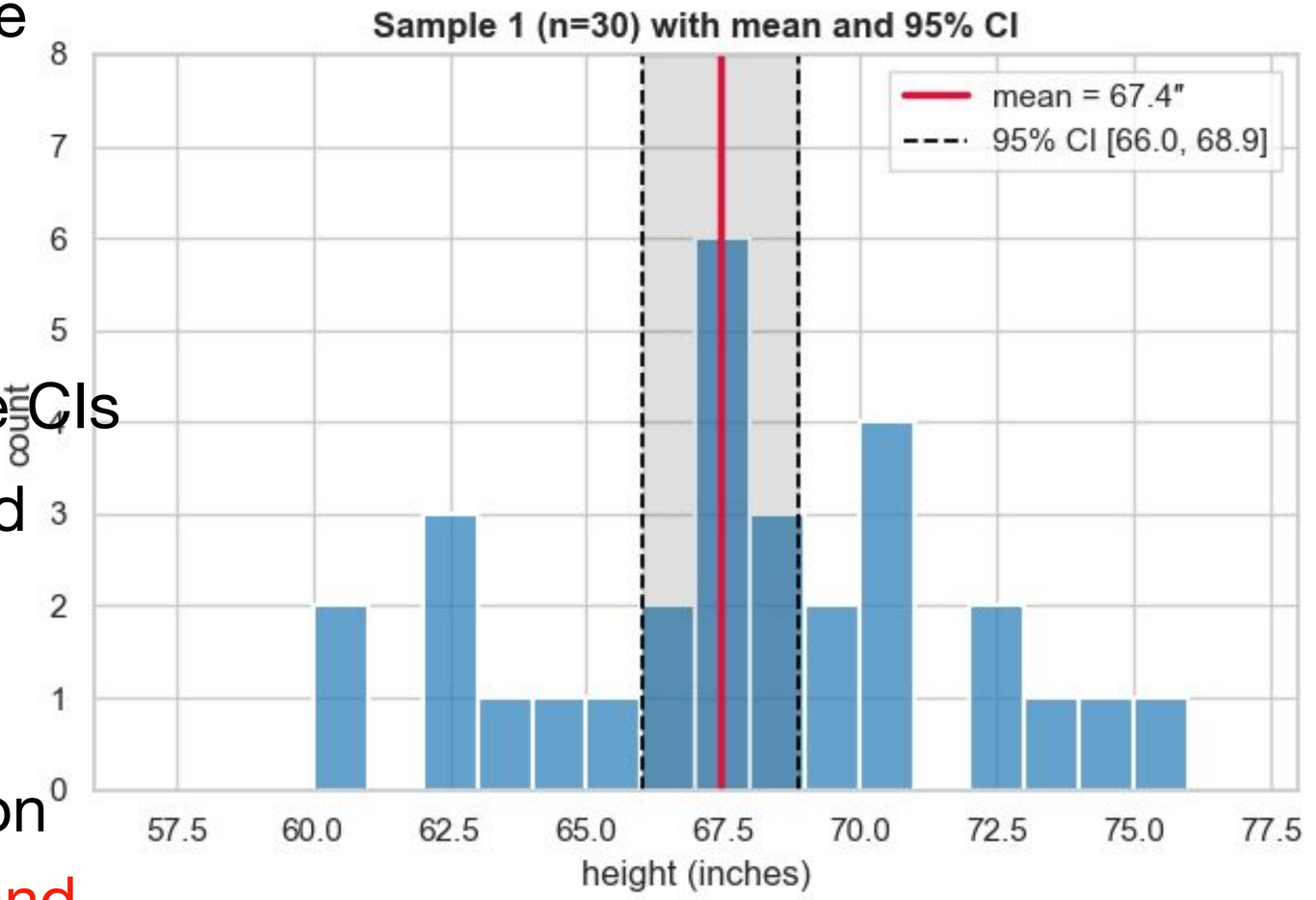
- The **standard error (SE)** quantifies how much a sample **statistic** deviates from the true population **parameter**
- ***Report:*** The sample mean height in COGS 9 is **66.3 inches** with **SE = 0.52 inches**
- This means that *across repeated random samples,* the sample mean height (**n=30**) differs from the true population mean height (**n=224**) by about **0.64 inches**

$$SE = \frac{s}{\sqrt{n}}$$



Quantifying uncertainty of a sample statistic

- A **confidence interval (CI)** uses the standard error to give a **range of plausible values** for the true parameter
- Across many samples, 95% of the CIs $[\text{mean} \pm 1.96 \times \text{SE(mean)}]$ would capture the true *population* mean height
- ***Report:*** We estimate the population mean height to be between 66.0 and 68.9 inches (95% CI)



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course code:
WSTAPX

Which of the following is true about a sample statistic (like the mean of heights)?

(**A**) It equals the population parameter

(**B**) The standard error (SE) is how much the sample statistic deviates from the true population parameter

(**C**) The sampling distribution is a distribution of the sample statistic over many samples

(**D**) All of the above

(**E**) B and C only

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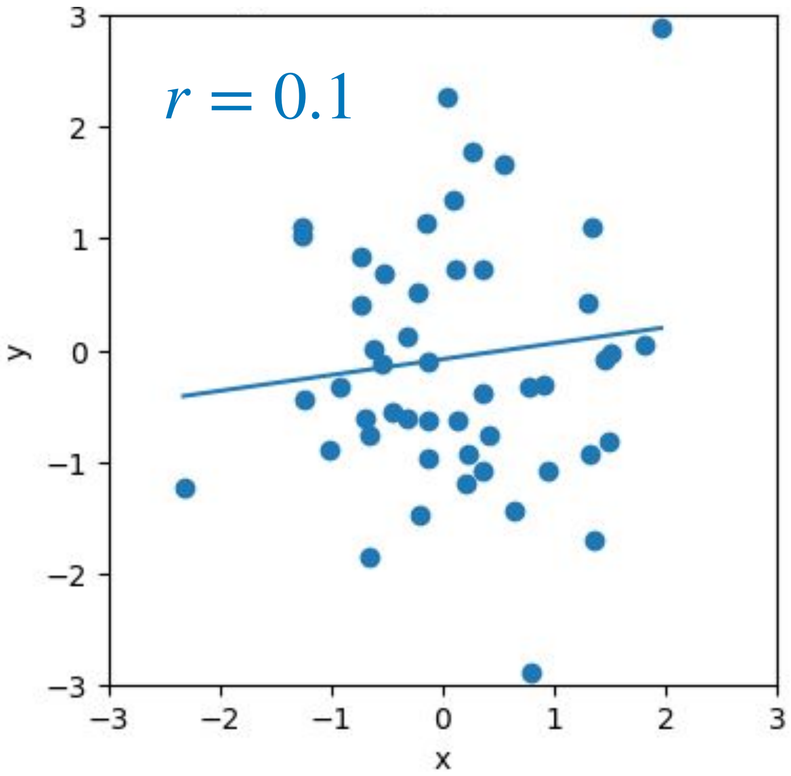
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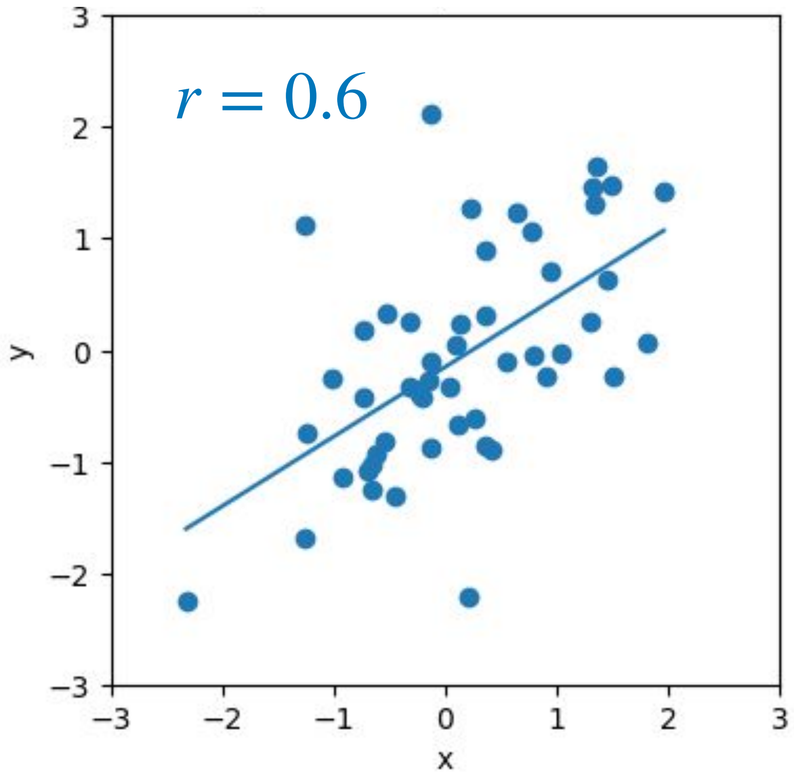
Bivariate statistical inference

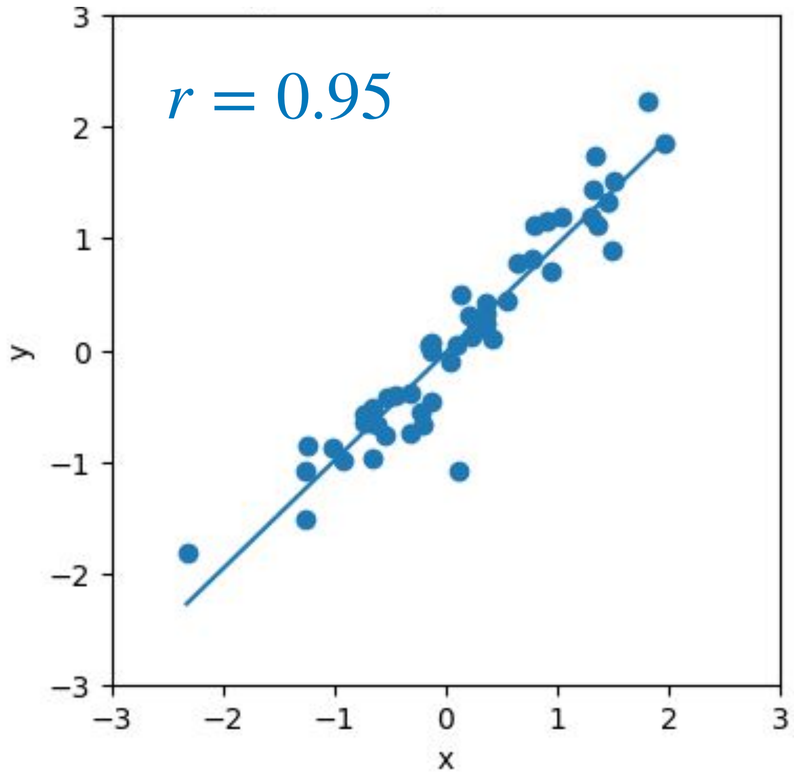
- **Bivariate inference:** draw conclusions about the relationship between two variables (comparison or association)
  - Quantitative ↔ quantitative: test association in values
    - **Inference tools:** correlation, slope
  - Categorical ↔ quantitative: difference in means across groups
    - **Inference tools:** t-test, ANOVA
  - Categorical ↔ categorical: test association or difference in proportions
    - **Inference tools:** chi-square test

Quantitative ↔ quantitative inference

- Pearson's correlation (r): a number from -1 to +1 that measures the **strength and direction** of a linear relationship between two quantitative variables
  - Stronger relationship (+ or -) → higher correlation

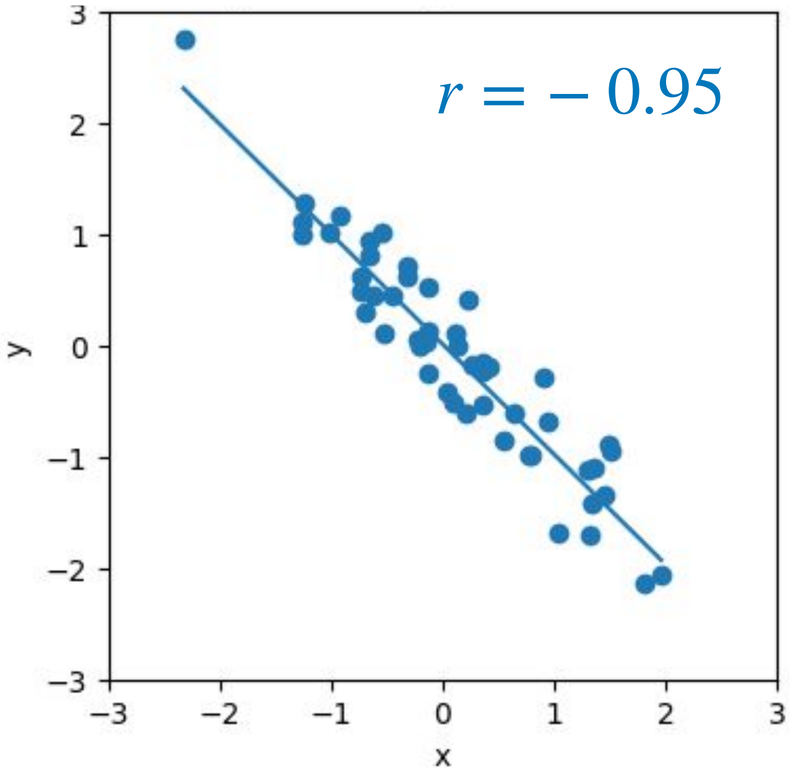


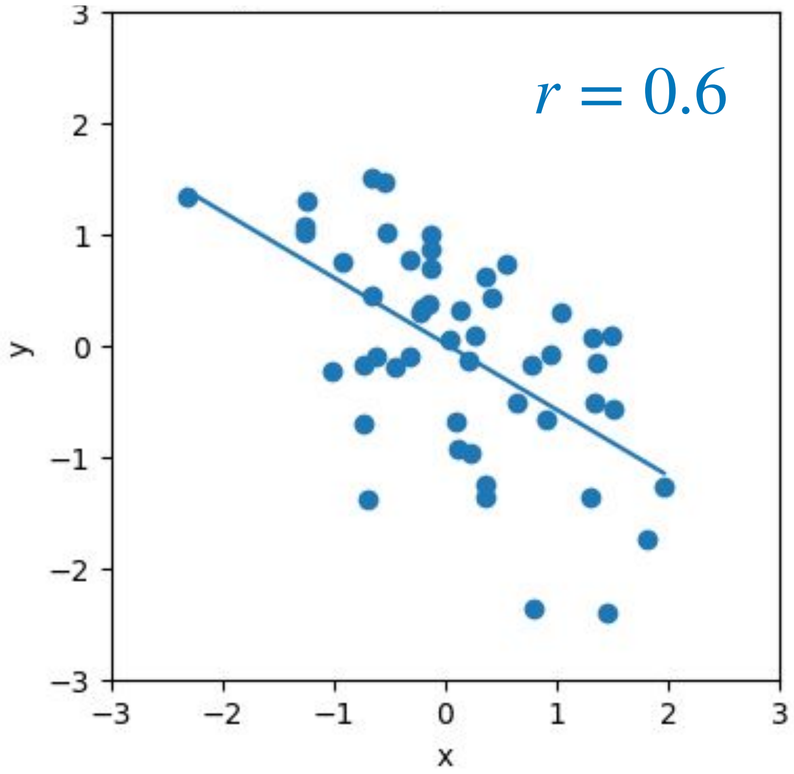


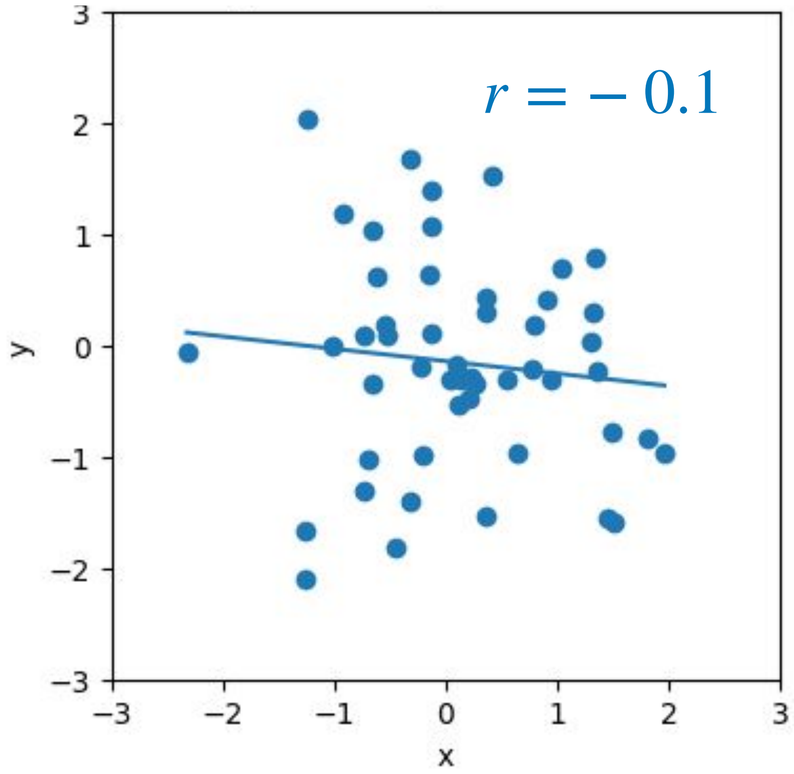


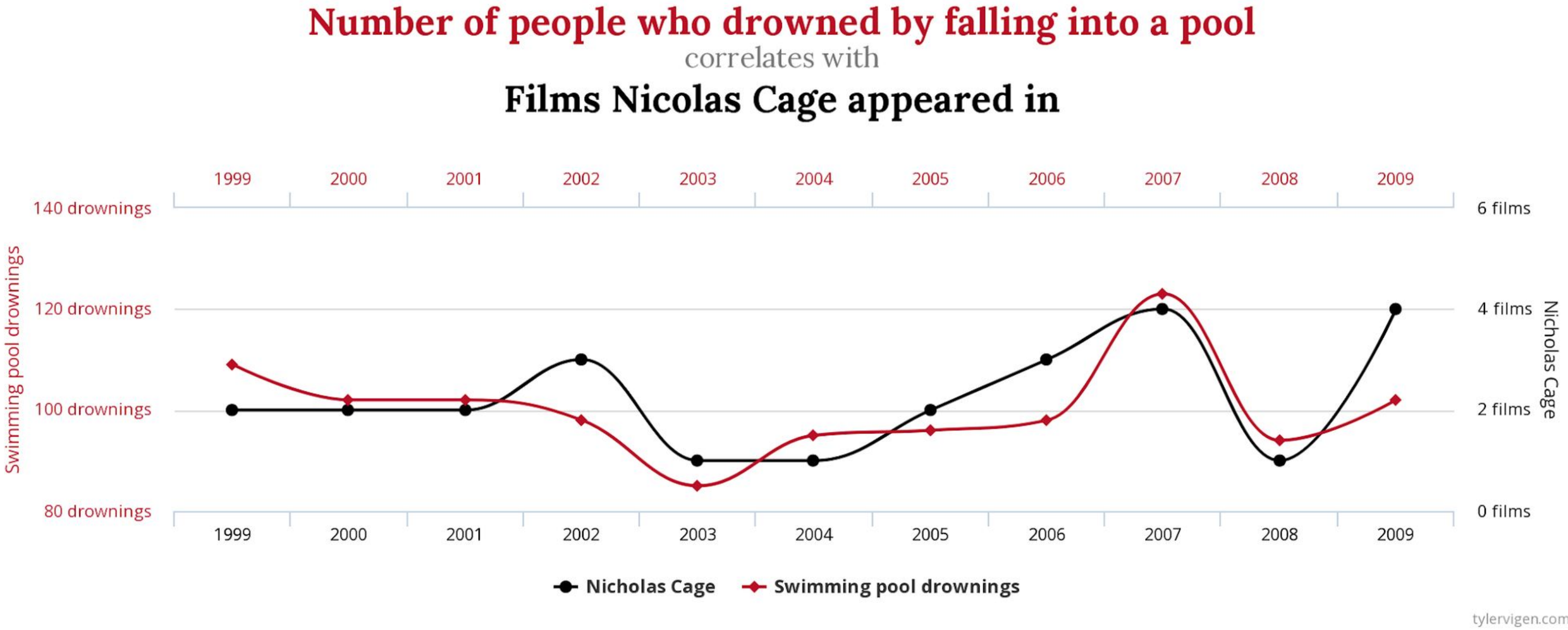
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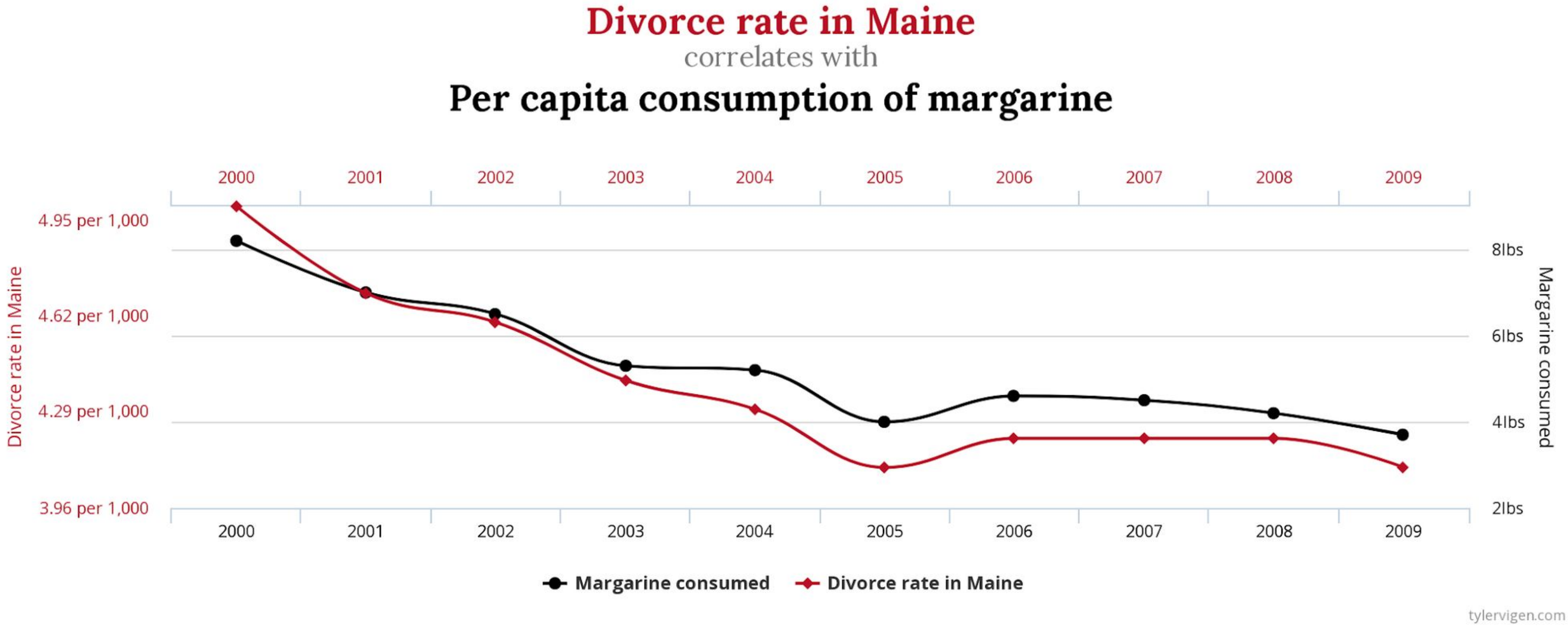








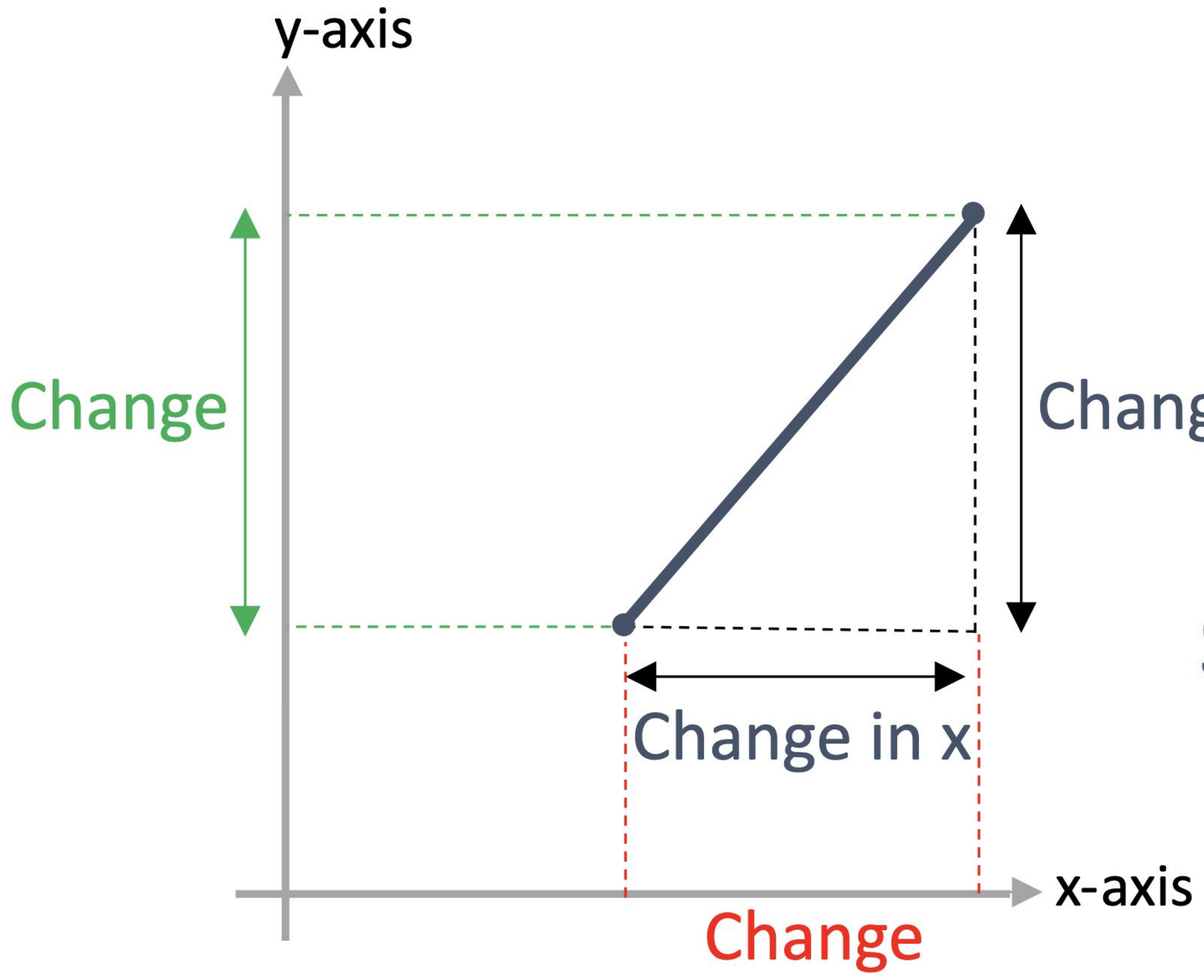
Correlation ≠ causation!!!



Correlation ≠ causation!!!

Quantitative ↔ quantitative inference

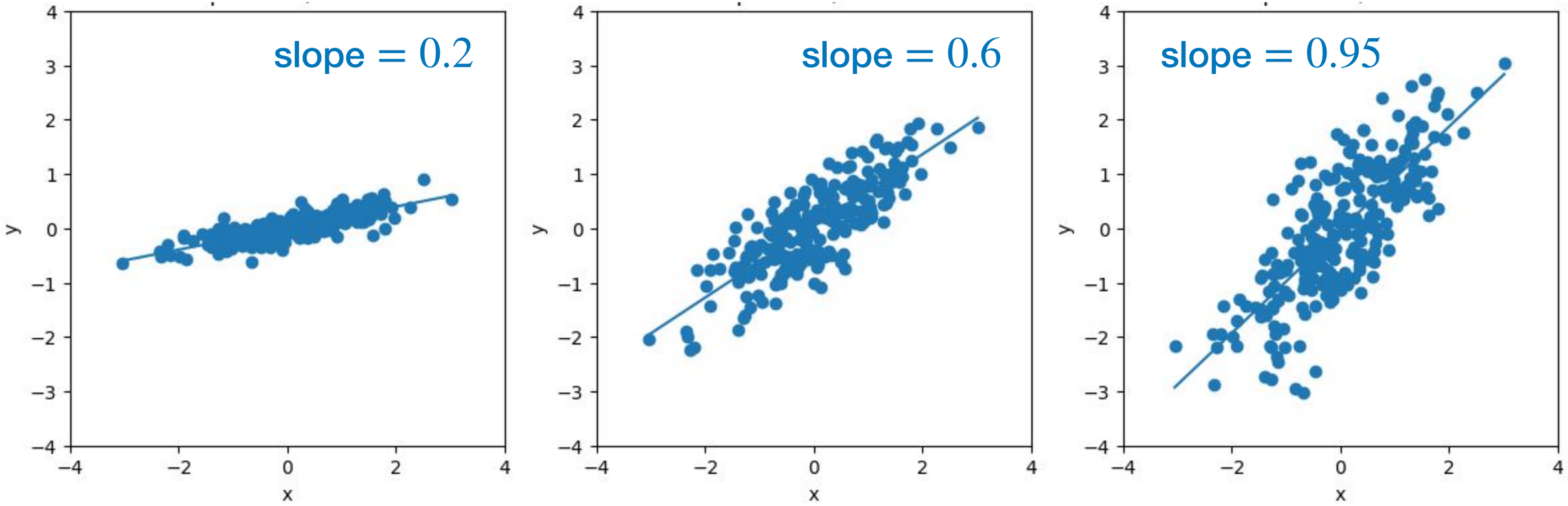
- Slope: the change in y that you expect for a 1-unit increase in x



$$\text{Slope} = \frac{\text{change in y}}{\text{change in x}}$$

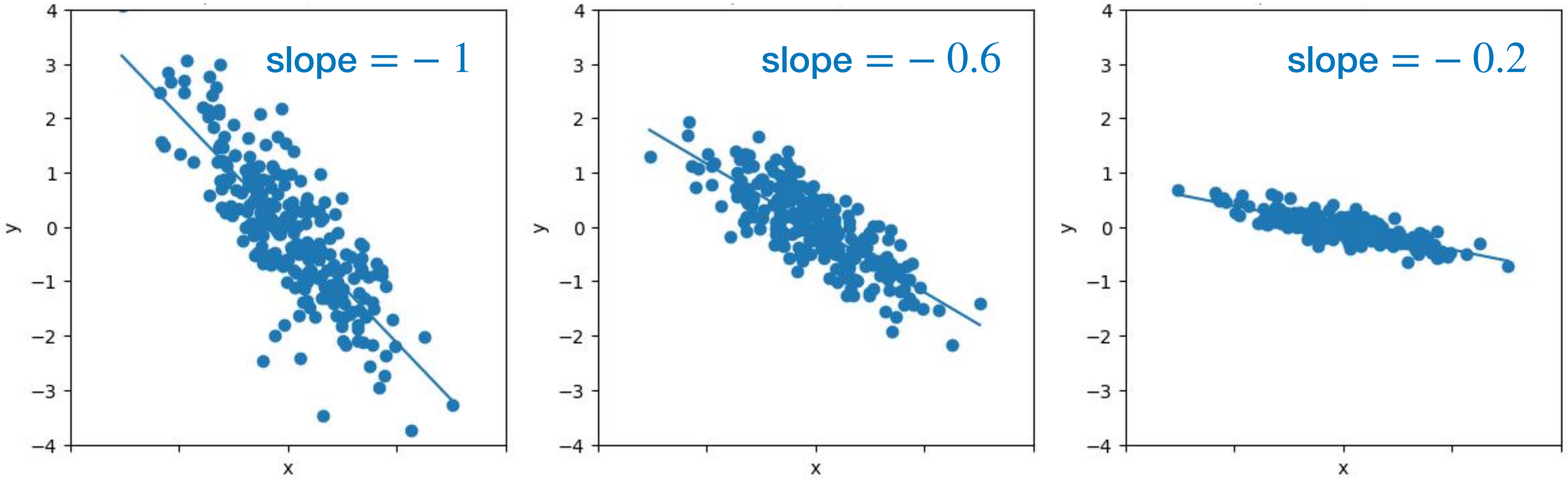
Quantitative ↔ quantitative inference

- Slope: the change in y that you expect for a 1-unit increase in x
- Correlation ≠ slope! All of these plots have the same correlation ($r = 0.8$)



Quantitative ↔ quantitative inference

- Slope: the change in y that you expect for a 1-unit increase in x
- Correlation ≠ slope! All of these plots have the same correlation ($r = -0.8$)



Slope is how much y changes with x

Correlation is how strongly x and y change *together*

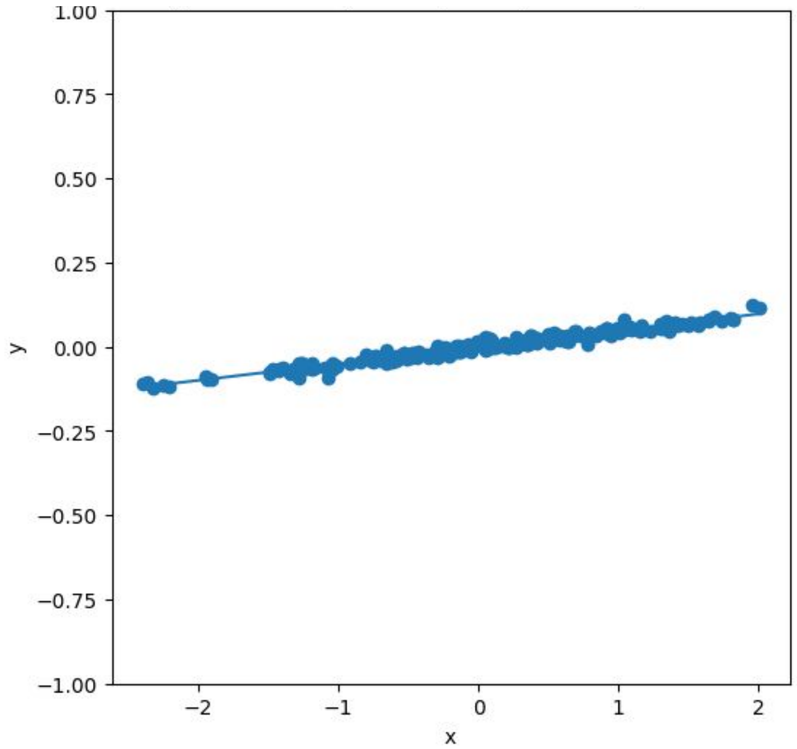
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course code:
WSTAPX

This plot has _____ and _____



(A) High correlation, low slope

(B) Low correlation, high slope

(C) High correlation, high slope

(D) I'm not sure / I'm confused

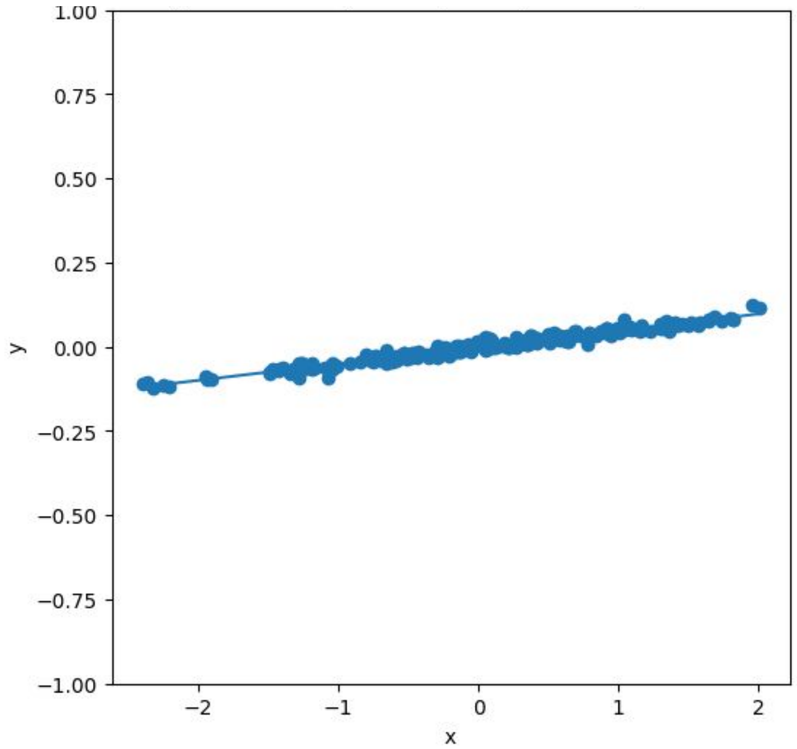
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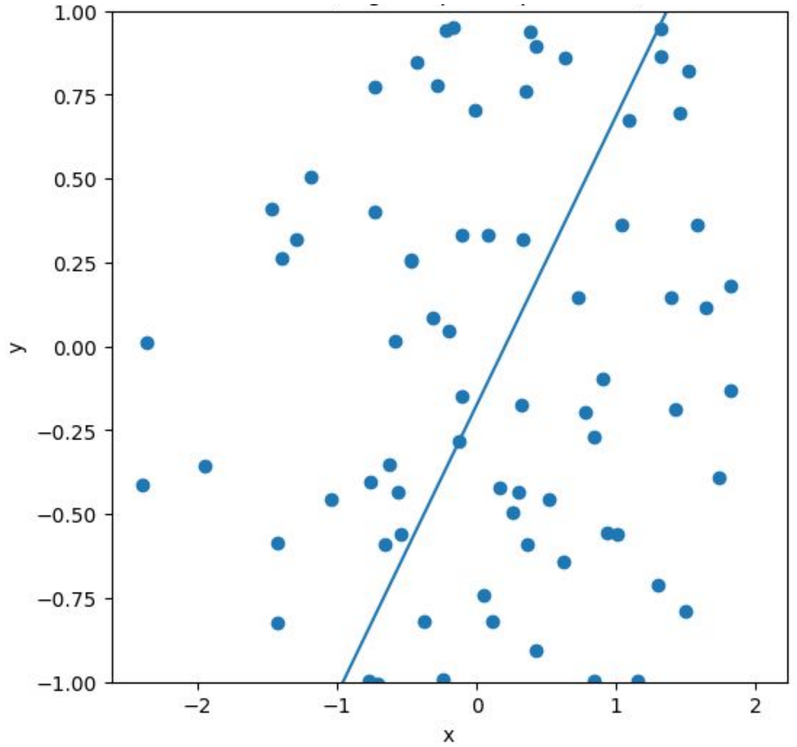
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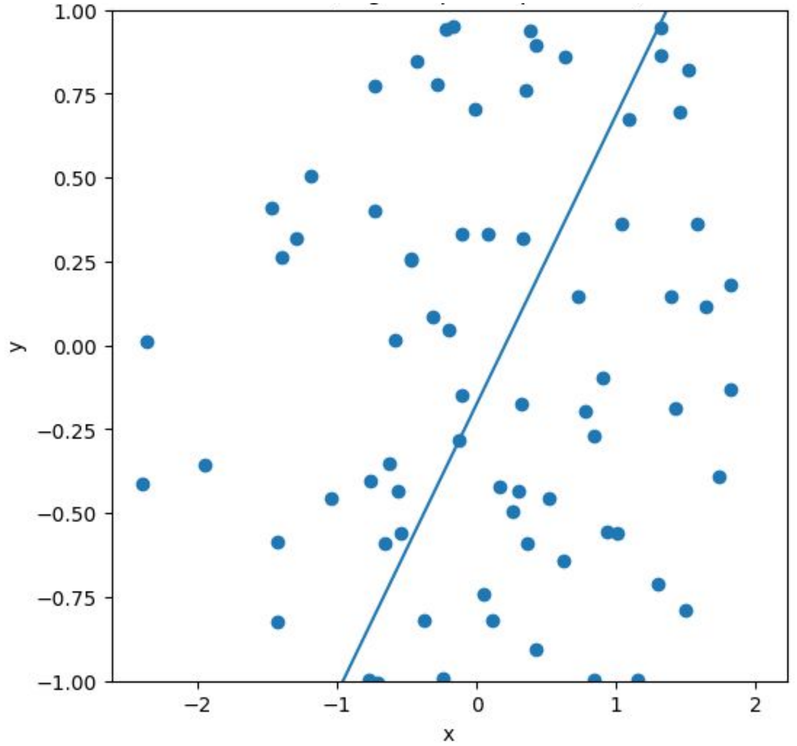
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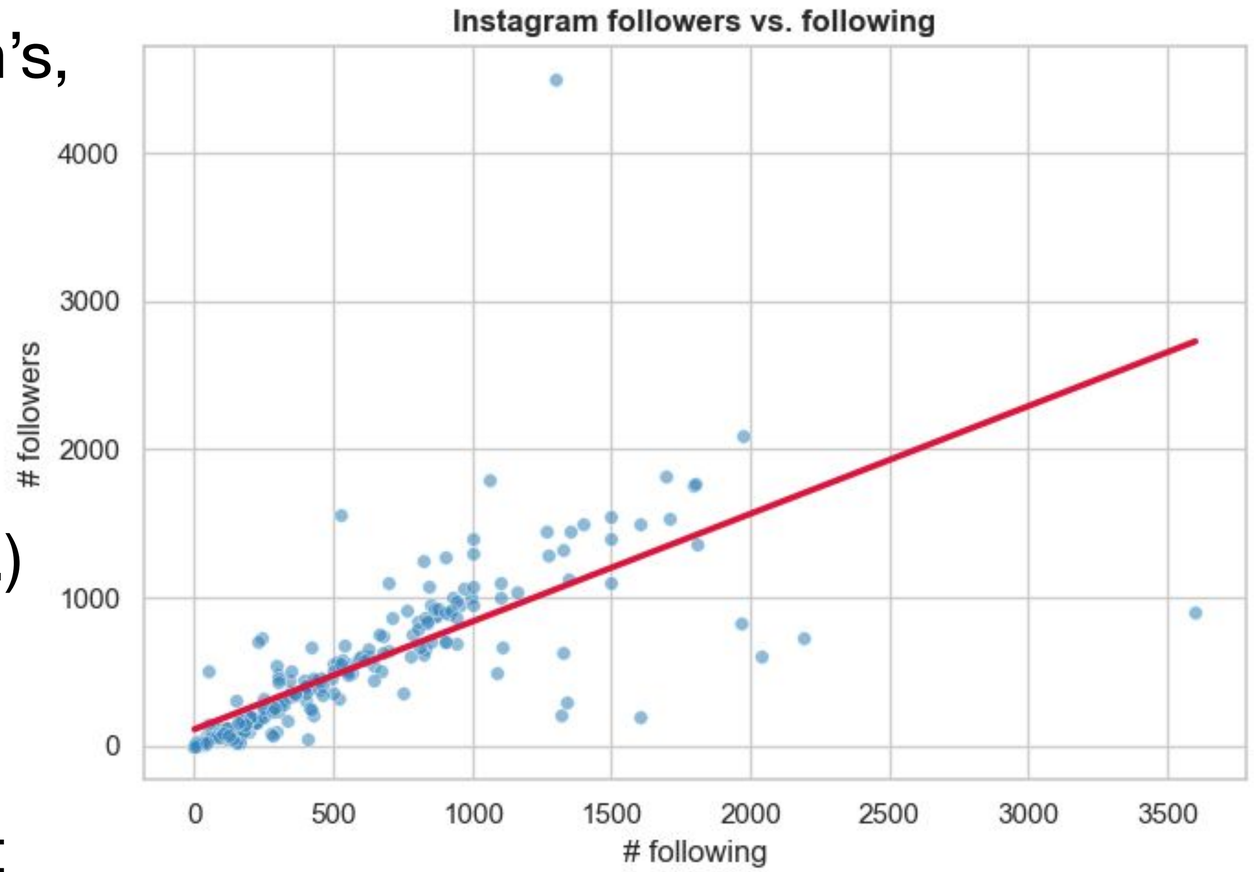
(B) Low correlation, high slope

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Quantitative ↔ quantitative inference

- Test association or difference in values
- **Inference tools:** correlation (Pearson's, Spearman's), slope
- ***Report:*** The Pearson's correlation between Instagram followers and following is $r = 0.73$ with a slope $= 0.72$ (coincidentally similar..)
- ***Conclude:*** On average, each additional account that you follow is associated with an *increase* of about 0.73 followers



Hypothesis testing

- **Hypothesis testing** is a statistical inference method used to test a claim
  - e.g., *Is there a correlation between # followers and # following?*
- Null hypothesis H_0: There is NO effect or difference between groups, and any observed difference is due to sampling variability (randomness across samples)
  - e.g., *The correlation between # followers and # following is 0*
- Alternative hypothesis H_A: There is an effect or difference between the groups
  - e.g., *The correlation between # followers and # following is different than 0*
- We can use statistical tests to determine whether to **reject the null** H_0

Hypothesis testing — directionality

- The question being tested can be directional:
  - **Directional:** *Do first-years in COGS 9 get **more sleep** than third-years?*
  - **Not directional:** *Does the number of sleep hours **differ** between first-years and third-years?*

Hypothesis testing — directionality

- Null hypothesis H_0: There is NO effect or difference between groups
  - **Directional:** *First-years get the* **same amount of sleep OR less sleep** *than third-years*
  - **Not directional:** *First-years and third-years get the* **same amount** *of sleep*
- Alternative hypotheses H_A: There is an effect (can be directional or not)
  - **Directional:** *First-years get* ***more sleep*** *than third-years*
  - **Not directional:** *The number of sleep hours* ***differs*** *between first- and third-years*
- The *directionality of* H_A in part determines what kind of statistical test you use (one-tailed vs. two-tailed t-test)

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course code:
WSTAPX

What is the null hypothesis H_0 for this data science question: *Do second or fourth-year students have higher GPAs?*

(**A**) Second-year students have a higher GPA than fourth-year students

(**B**) Fourth-year students have a higher GPA than second-year students

(**C**) Second- and fourth-year students have the same GPA

(**D**) Second- and fourth-year students have different GPAs

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Hypothesis testing and p-values

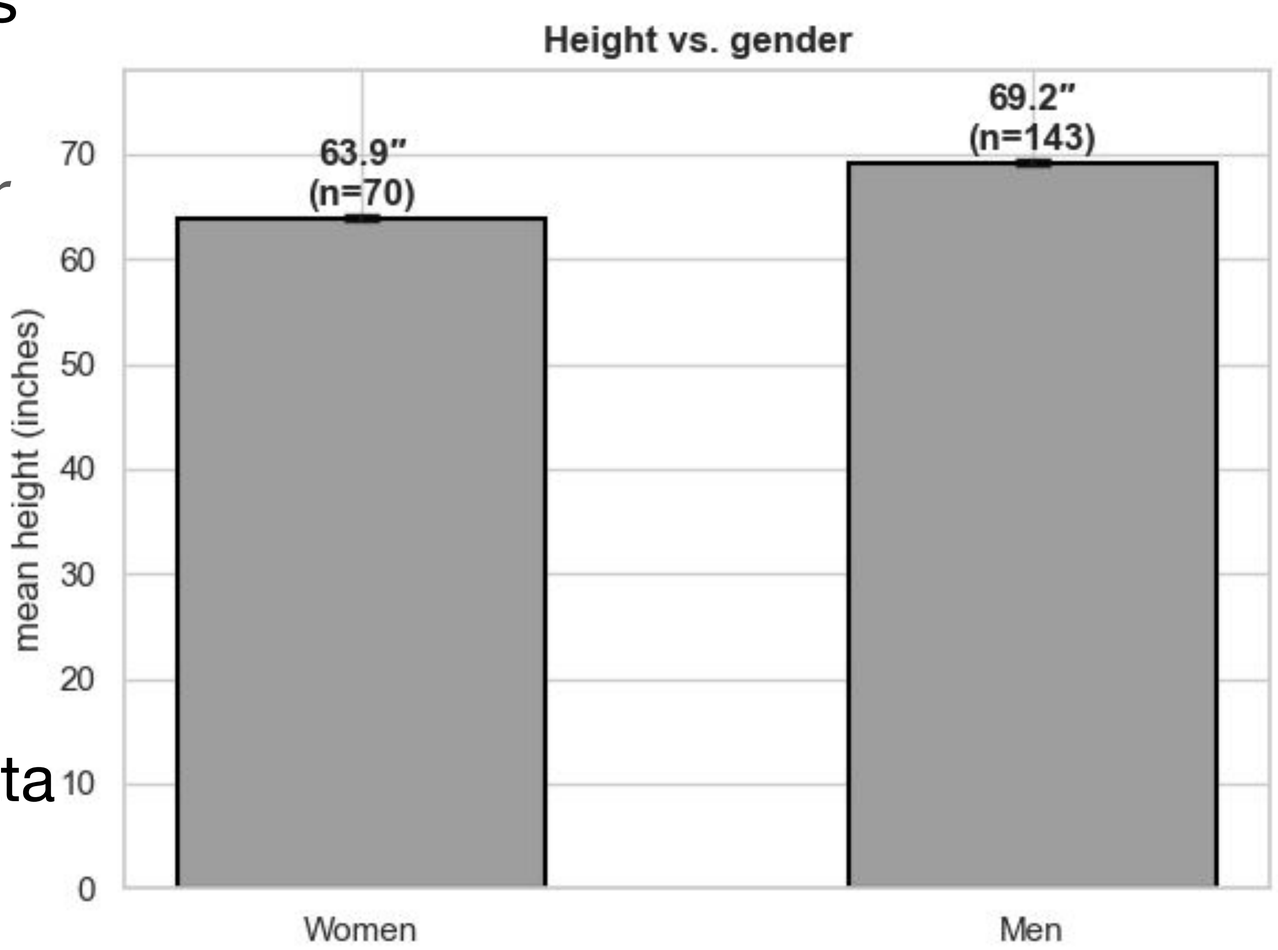
- How do you know if you can reject the null hypothesis H_0?
- p-value is a statistical metric that helps determine if observed results are **actually meaningful** or due to sampling variability (randomness)
  - p-value: the probability of getting the observed results by chance alone
  - If p is small (e.g., $p < 0.05$)

    → small probability that you would get those results just by chance

    → your results are likely *significant* (reject H_0 , accept H_A)

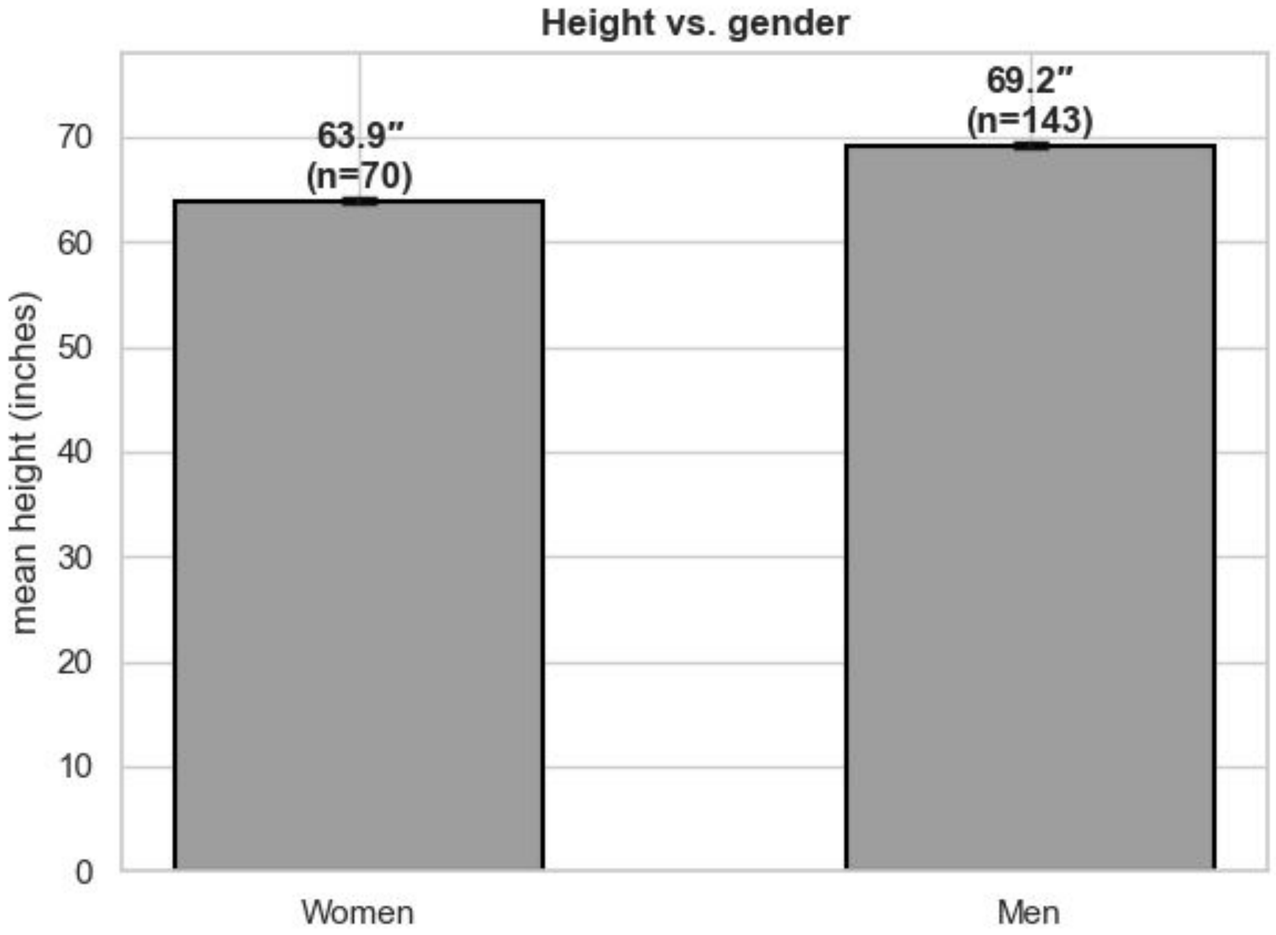
Categorical ↔ quantitative inference

- T-test tests for differences in means across 2 groups
  - e.g., *Does height differ by gender among COGS 9 students?*
- T-test assumptions (this is why univariate EDA is important!)
  1. Values are continuous
  2. Values are normally distributed
  3. The variance (spread) of the data in both groups should be approximately equal
  4. No outliers



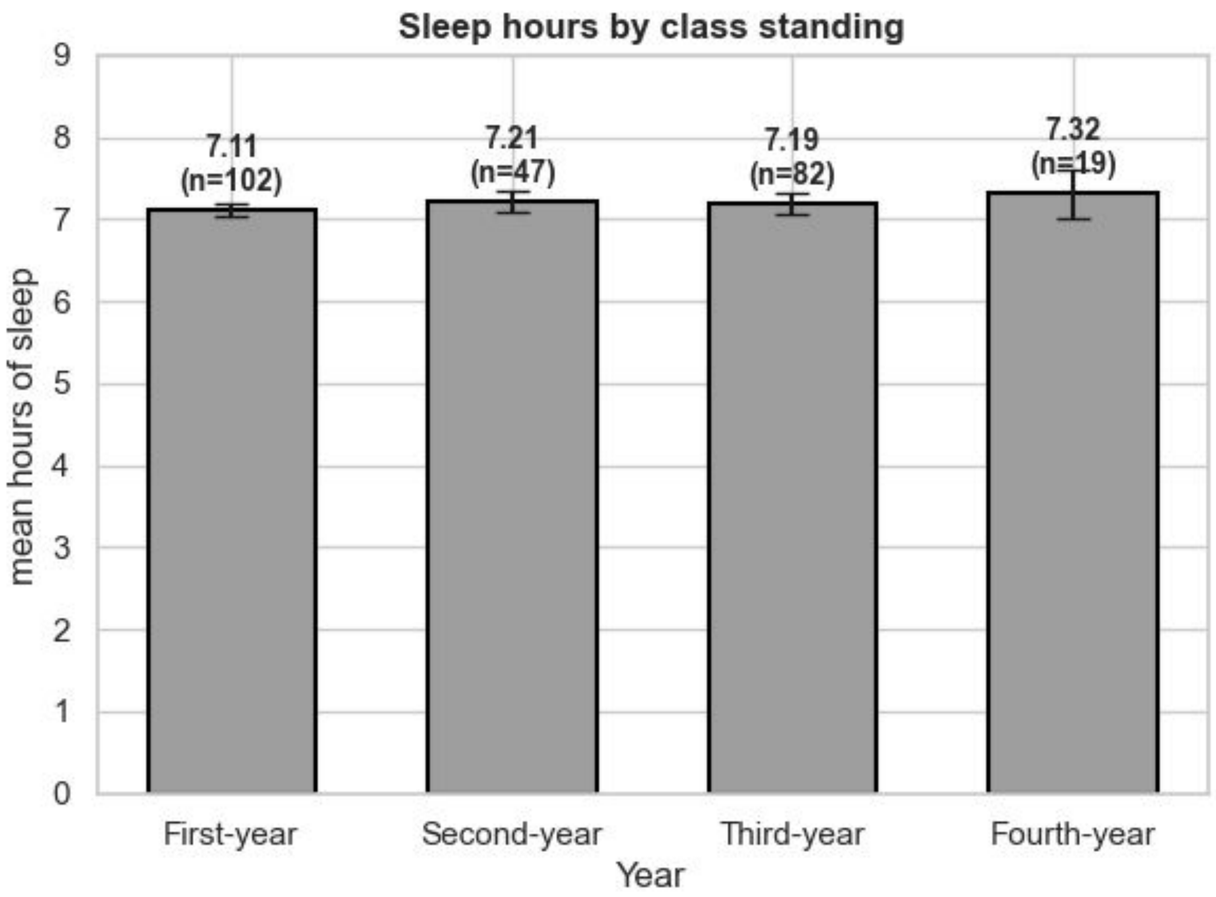
Categorical ↔ quantitative inference

- **T-test** tests for differences in means across 2 groups
  - e.g., *Does height differ by gender among COGS 9 students?*
- ***Report:*** $t = 14.4, p < .001$
- ***p-value interpretation:*** The probability of seeing this difference in means *by random chance alone* is less than 1 in 1000 → very small!
- ***Conclude:*** The difference in age is *significant*—coffee drinkers are older than tea/matcha drinkers!



Categorical ↔ quantitative inference

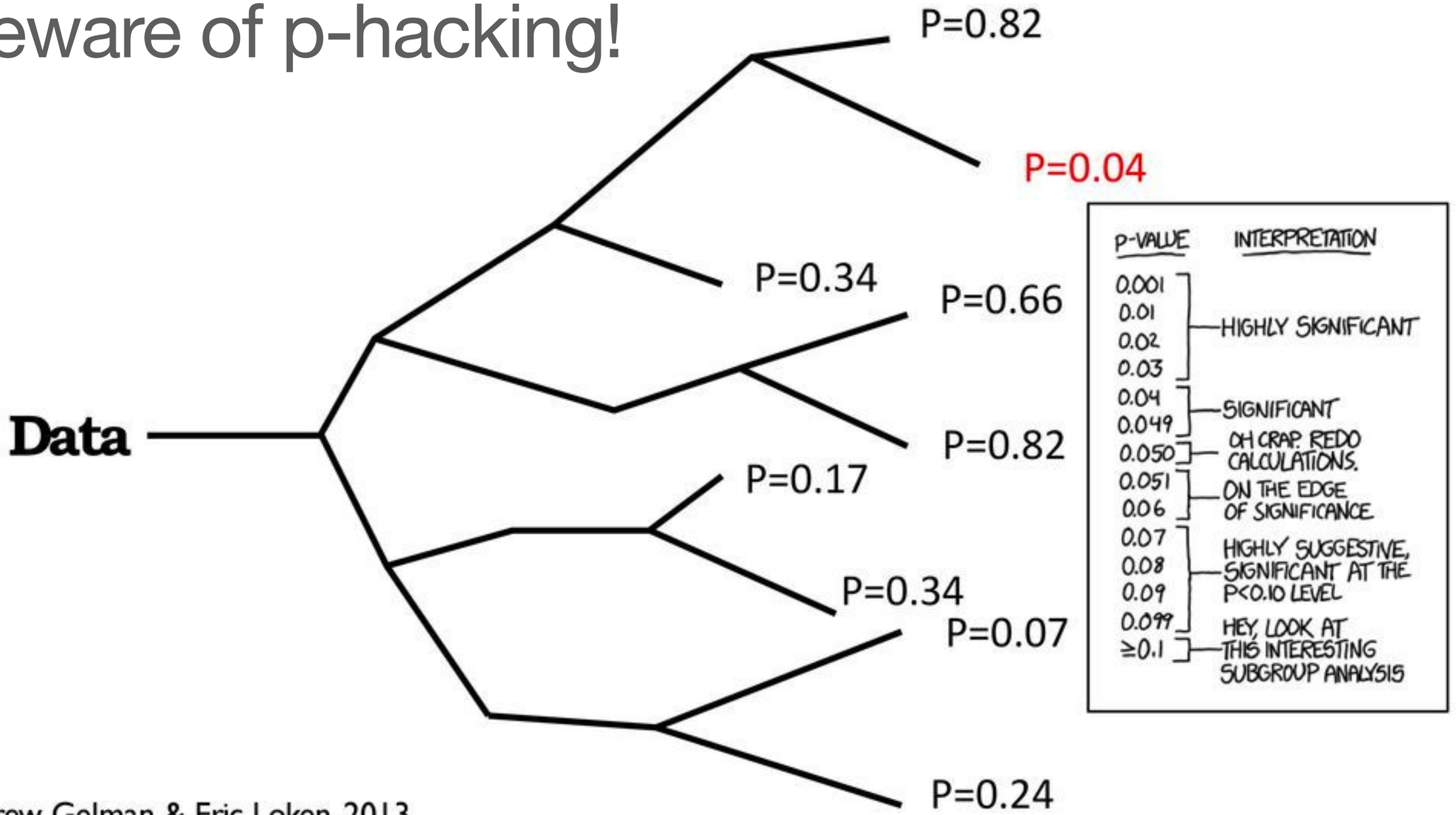
- **ANOVA (Analysis of Variance)** tests for differences in means across 3 or more groups
  - e.g., *Do sleep hours differ across years in COGS 9?*
- ANOVA assumptions: same as t-test
- ***Report:*** $F = 0.328, p = 0.805$
- ***p-value interpretation:*** The probability of seeing a difference in means *by random chance alone* is about 80 in 100 → fairly large
- ***Conclude:*** The difference in sleep hours across groups is NOT significant



Beware of p-hacking!

- p-hacking is manipulating data or analyses until a desired statistically significant result appears (usually $p < 0.05$)
  - Stopping data collection early because you reached $p < 0.05$
  - Excluding data to get $p < 0.05$
  - Only reporting analyses with $p < 0.05$
  - Manipulating or transforming the data to get $p < 0.05$
- DON'T p-hack! It ultimately leads to unreliable findings and poor science

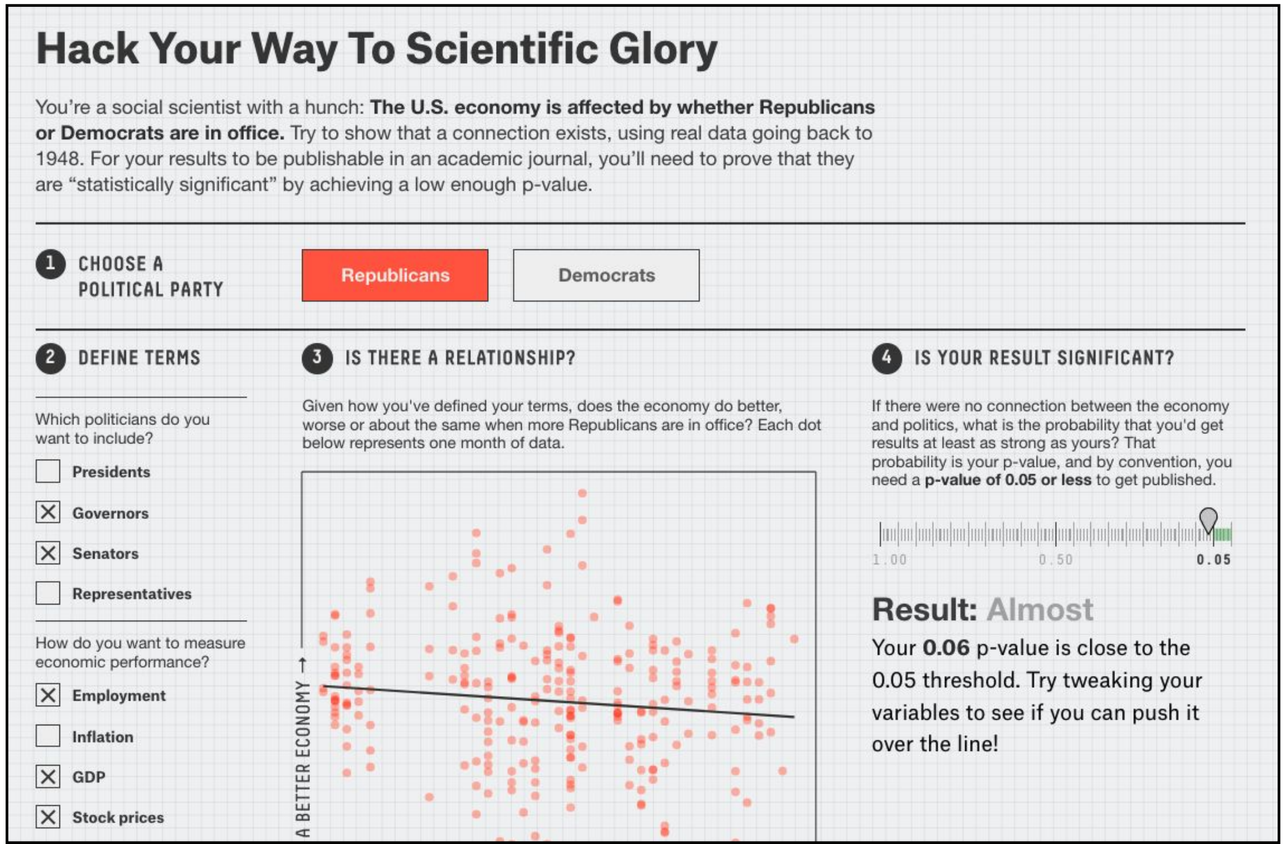
Beware of p-hacking!



Andrew Gelman & Eric Loken, 2013

Assignment 2: p-hacking

- Due Wednesday Week 8
- You will be using an interactive web tool to perform p-hacking yourself to understand how it can happen during data analysis!



Hack Your Way To Scientific Glory

You're a social scientist with a hunch: **The U.S. economy is affected by whether Republicans or Democrats are in office.** Try to show that a connection exists, using real data going back to 1948. For your results to be publishable in an academic journal, you'll need to prove that they are "statistically significant" by achieving a low enough p-value.

1 CHOOSE A POLITICAL PARTY

Republicans | Democrats

2 DEFINE TERMS

Which politicians do you want to include?

- ☐ Presidents
- ☒ Governors
- ☒ Senators
- ☐ Representatives

How do you want to measure economic performance?

- ☒ Employment
- ☐ Inflation
- ☒ GDP
- ☒ Stock prices

3 IS THERE A RELATIONSHIP?

Given how you've defined your terms, does the economy do better, worse or about the same when more Republicans are in office? Each dot below represents one month of data.

A BETTER ECONOMY ↑

4 IS YOUR RESULT SIGNIFICANT?

If there were no connection between the economy and politics, what is the probability that you'd get results at least as strong as yours? That probability is your p-value, and by convention, you need a **p-value of 0.05 or less** to get published.

1.00 0.50 0.05

Result: Almost

Your **0.06** p-value is close to the 0.05 threshold. Try tweaking your variables to see if you can push it over the line!

COGS 9: Introduction to Data Science
Assignment #2: p-hacking (9pts)
Submit through Gradescope individually

First Name	Last Name	PID

Introduction

For most studies, "statistical significance" is defined as a p-value less than 0.05. In plain terms, that means the observed difference between two groups would be expected to occur by chance less than about 1 in 20 times, which is often treated as a fairly high bar for proving that the difference is "real."

But what if a researcher runs 20 different comparisons and reports only the one that looks "significant"? Or what if a trend starts to appear in the data and samples are dropped or added until the p-value hits the magic cutoff of 0.05? If these decisions aren't disclosed, and if the raw data aren't available for independent checks, readers may be misled. This is the problem known as *p-hacking*.

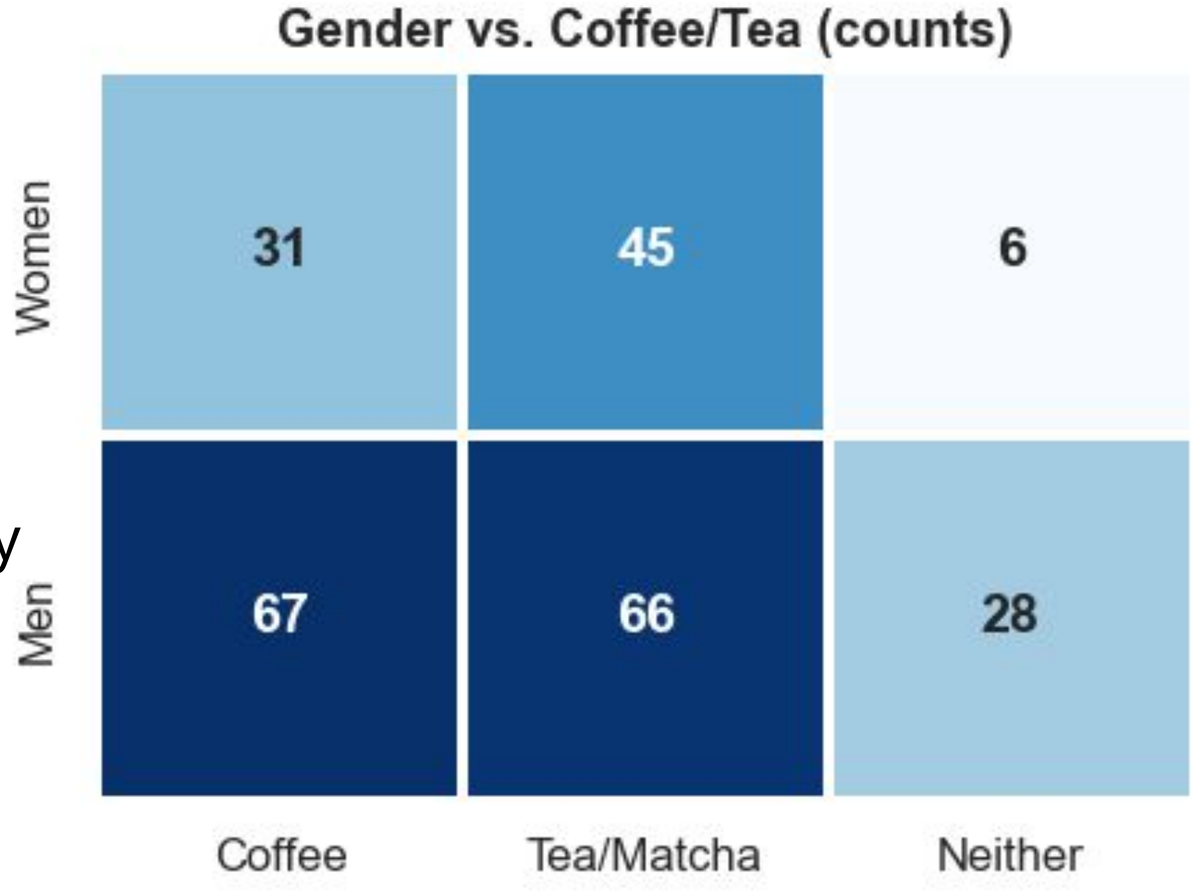
What you'll be doing

In this assignment, you'll use an interactive tool to explore the relationship between political parties and economic outcomes as an example case study of how statistical analysis choices can dramatically affect your results. You will:

1. **Generate a hypothesis** about the relationship between political parties and the economy
2. **Test your hypothesis** in an interactive web tool and document the results

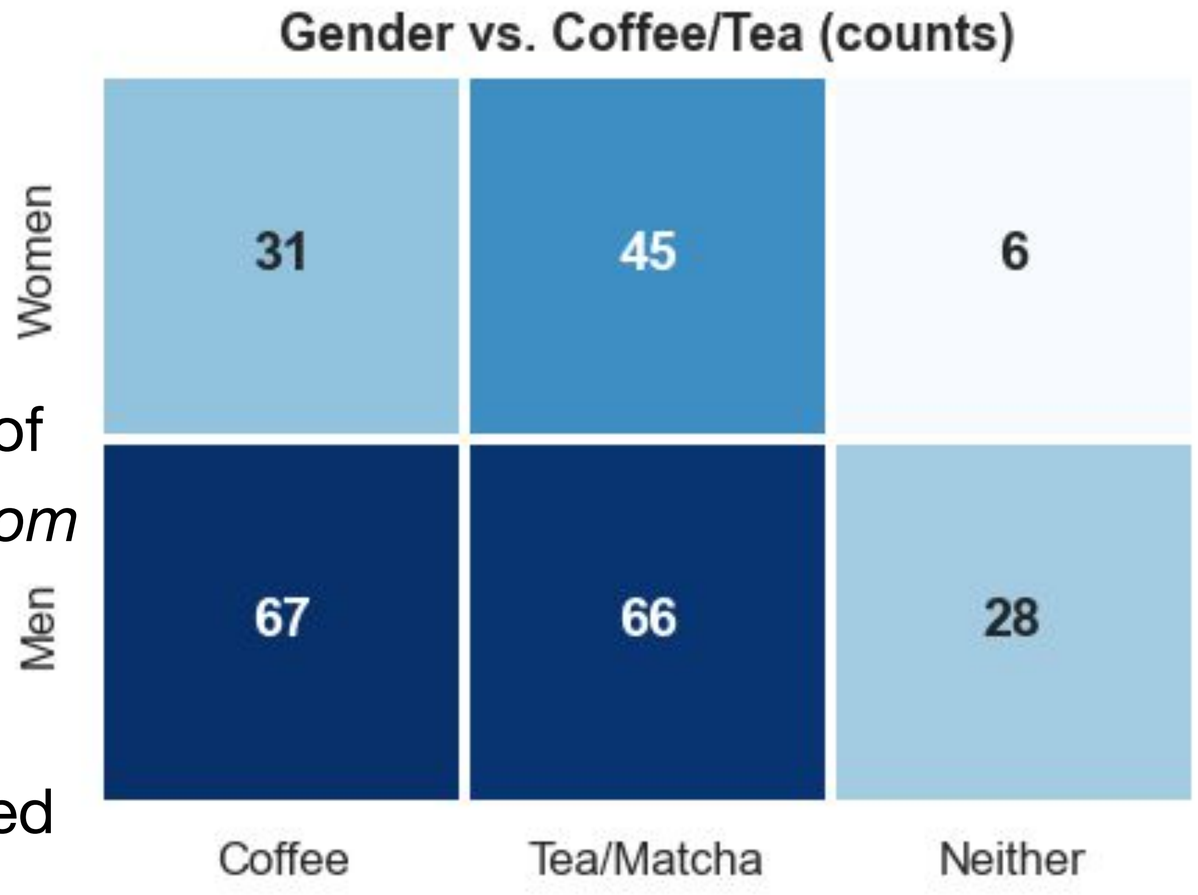
Categorical ↔ categorical inference

- **Chi-squared (χ^2) test** tests for association or difference in proportions
  - e.g., *Do drink preferences differ by gender?*
- Chi-squared test assumptions:
  1. Values are categorical counts
  2. At least >5 counts in each category
  3. Each data point is only in 1 category



Categorical ↔ categorical inference

- **Chi-squared (χ^2) test** tests for association or difference in proportions
  - e.g., *Do drink preferences differ by gender?*
- ***Report:*** $\chi^2 = 6.43$, $p = 0.04$
- ***p-value interpretation:*** The probability of seeing this difference in means *by random chance alone* is about 4 in 100 → very small!
- ***Conclude:*** Drink preference *is* associated with gender—women like tea/matcha more than men, and more men like neither



Multivariate statistical inference

- **Multivariate inference:** drawing conclusions about relationships between variables while accounting for other variables (confounding and interaction effects)
  - **Inference tools:** multiple regression (linear/logistic), ANCOVA, etc.
- *+ many further topics to explore in your statistics class*!

Discussion (~5 mins)

Groups of 2-3:

Data science question: *Do first-year students attend lecture more than non-first-year students in Week 6 of COGS 9, Fall 2026?*

1. What is the null hypothesis?
2. What is the alternative hypothesis?
3. What kind of **EDA plot** would you make to explore this question?
4. What kind of **statistical test** would you use to test this question?

Takeaways

- The goal of statistical inference is to draw conclusions about a population from a sample of data
- Let your hypothesis guide your choice of EDA and statistical test
  - Are you…
    - using a **sample** to estimate a **population** parameter?
    - testing an **association** between two variables?
    - **comparing** two groups or categories?
    - exploring the relationship between **multiple variables**?